Data & AI

3. Function Approximation

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Today

jdasam.github.io/mas1004/demos/lecture03-regression-search

- Fit a line to data by hand and measure how well it fits
- Let a computer search for the best fit
- Find where the search stops being possible

One input, one output

Five Datasets

One number in, one number out

Input x	Output y	Source
Floor area of a studio (m^2)	Monthly rent (10,000 KRW)	Synthetic, studios near Sinchon
Daily high temperature (°C)	Iced americanos sold at a campus cafe	Synthetic
Phone screen time (hours a day)	Sleep (hours a night)	Synthetic
Parents' height (cm)	Adult child's height (cm)	Galton (1886), 928 children
Study hours a week	GPA (out of 4.3)	Synthetic

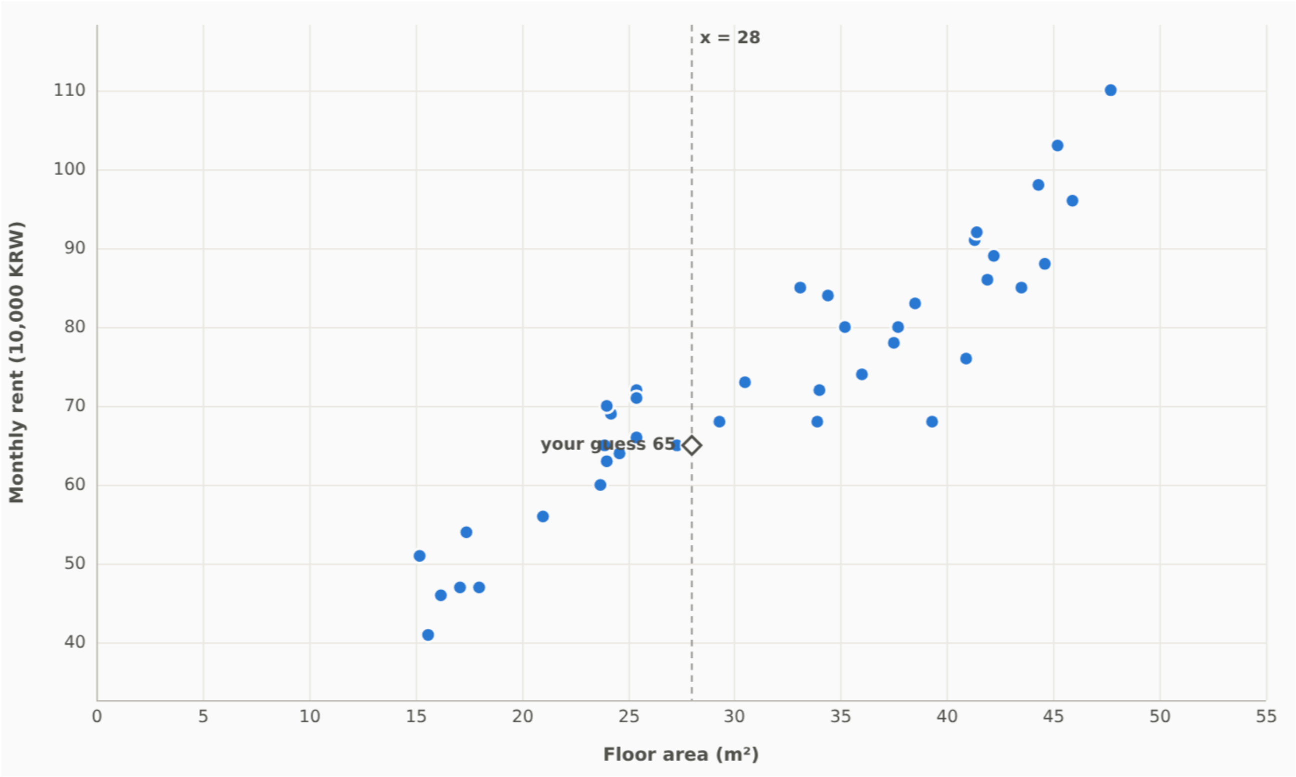
Data as Points

Studio floor area and monthly rent

Studio floor area and monthly rent Only the points. No function yet.

Rent | Cafe | Sleep | Galton 1886

Data



● one sample ┅ the x we are asked about

A 28 m² studio near campus. What monthly rent should you expect?

Your guess by eye: 65

× 10,000 KRW

40 samples. Each point is one listing: x = floor area (m²), y = monthly rent (10,000 krw). Synthetic data, fixed seed. Hover a point to read its values.

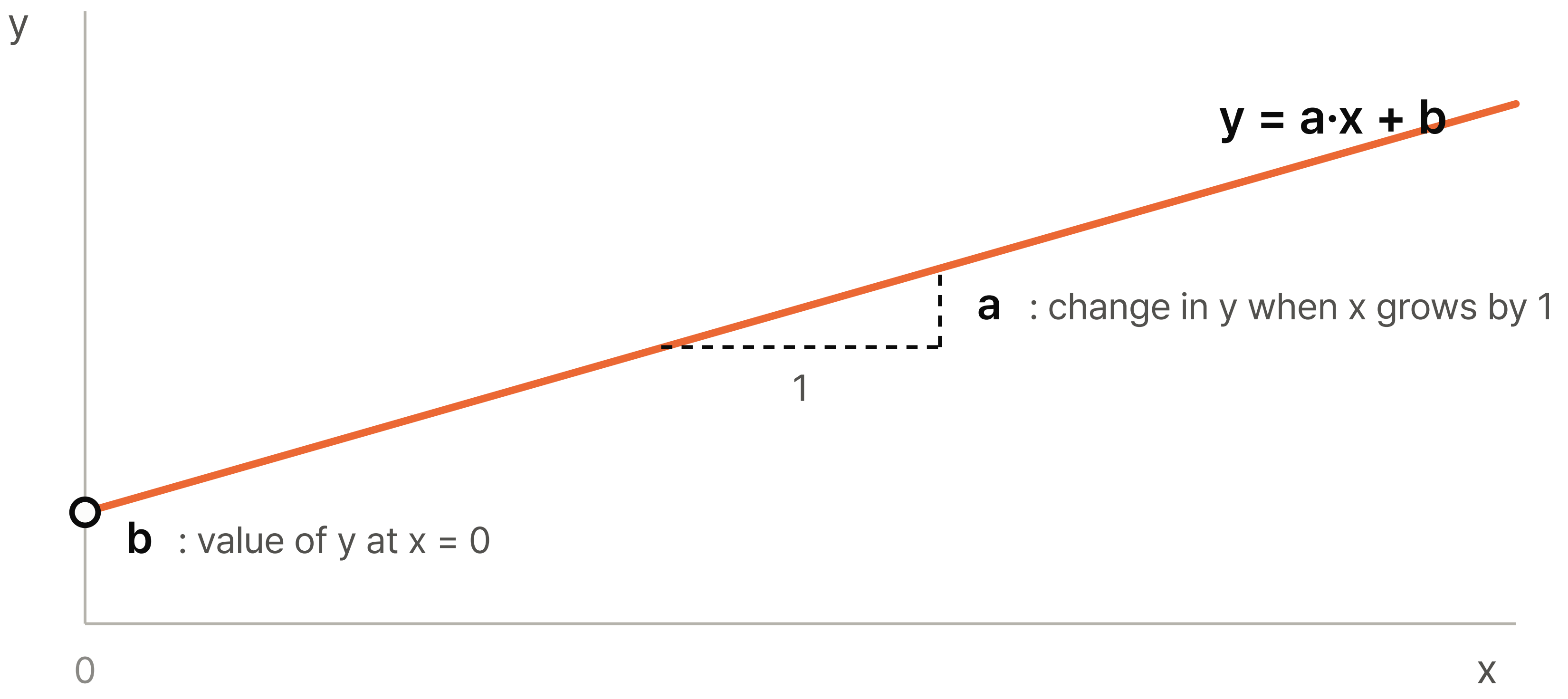
Two Things We Ask of the Data

- Predict a value: the rent of a 28 m^2 studio that is not in the data
- Read the trend: how much the rent rises per square meter

A line through the points

The Model

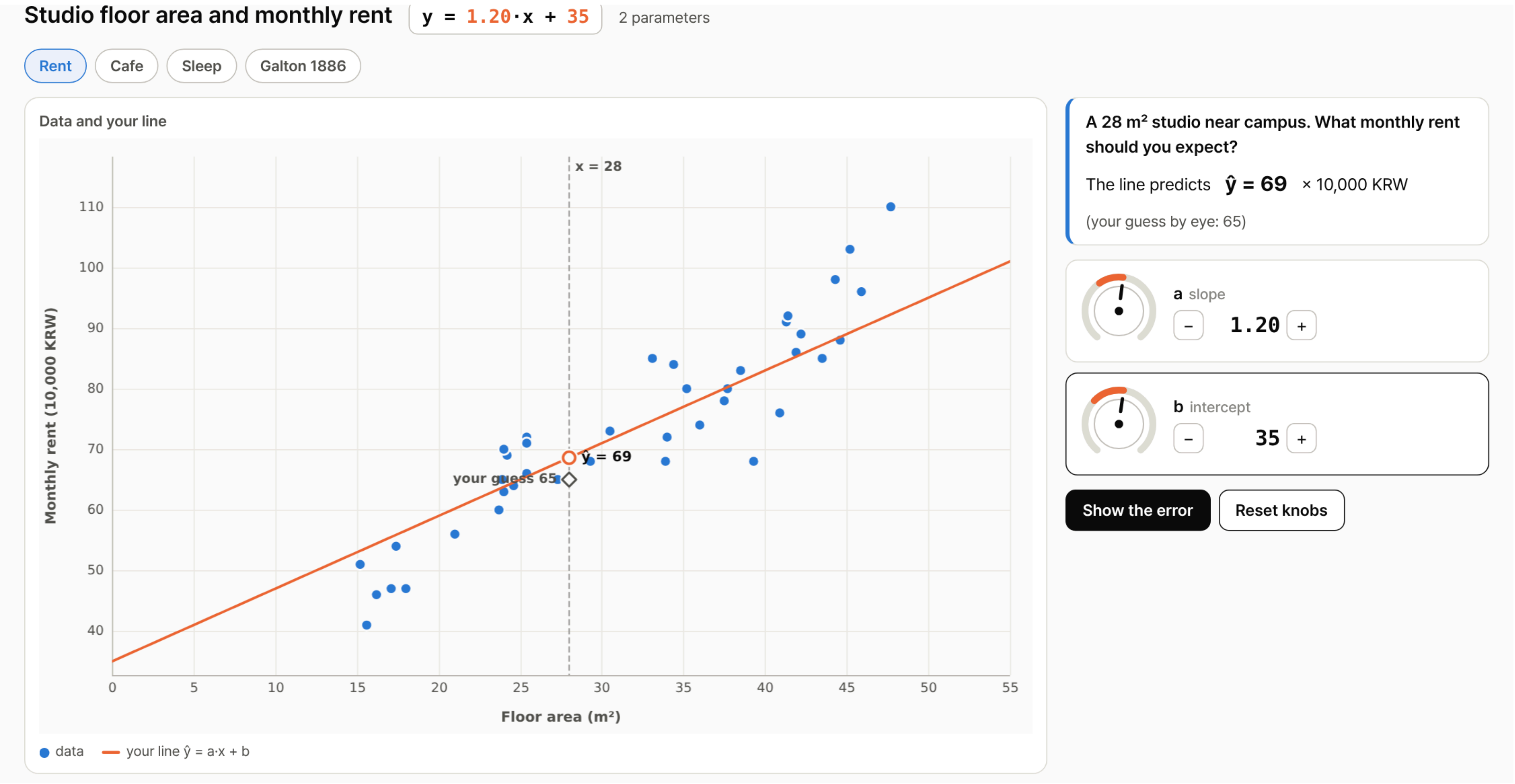
$y = a \cdot x + b$



- a and b are the parameters: two numbers we choose

Fitting by Hand

Demo, tab 2



Which Line Is Better?

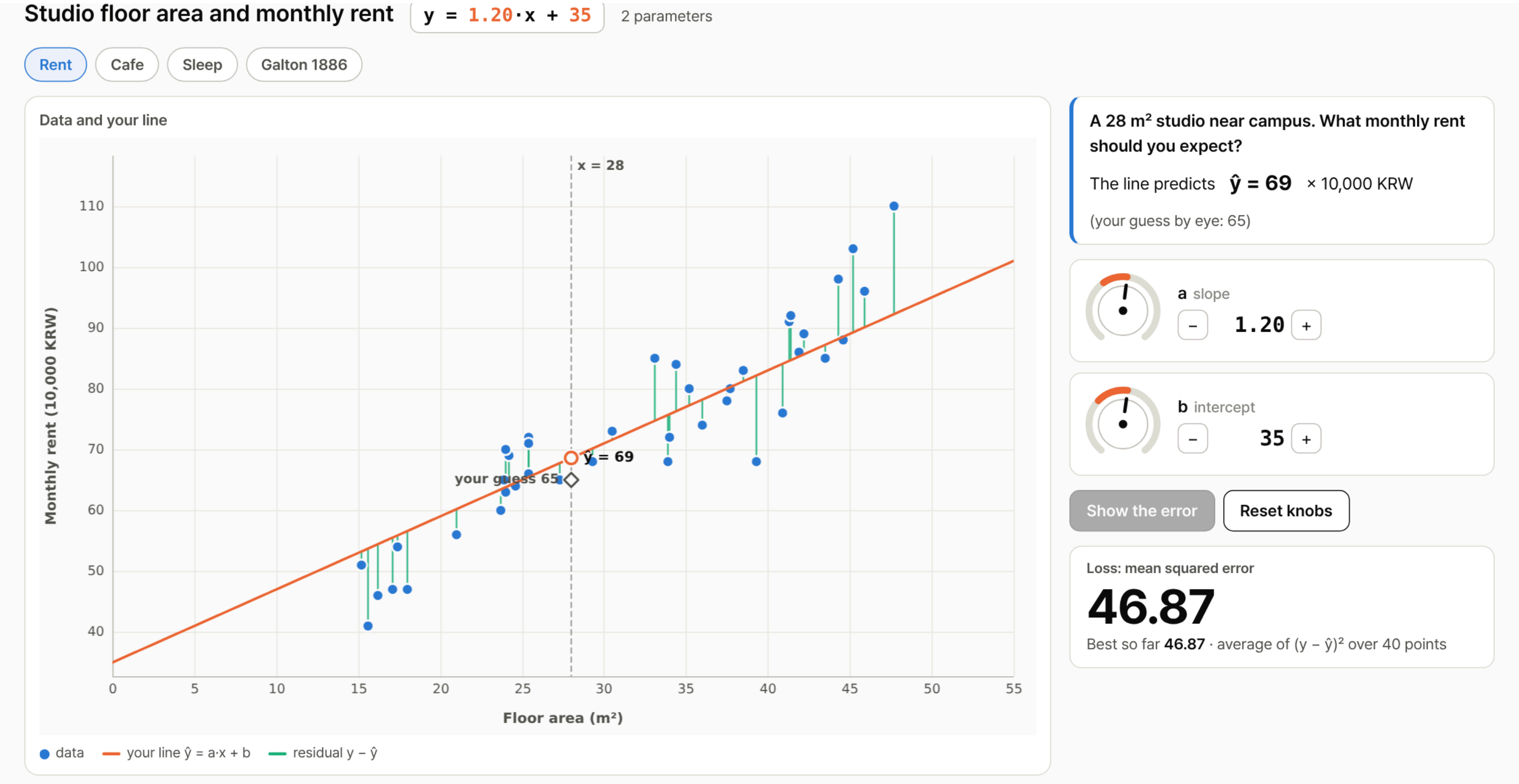
- Residual of one point: $y_i - \hat{y}_i$, the vertical distance from the point to the line
- Loss: the mean of the squared residuals

$$\text{Loss} = \frac{1}{n} \sum_{i = 1 \ldots n} (y_i - \hat{y}_i)^2$$

- One number for the whole line. Smaller is better

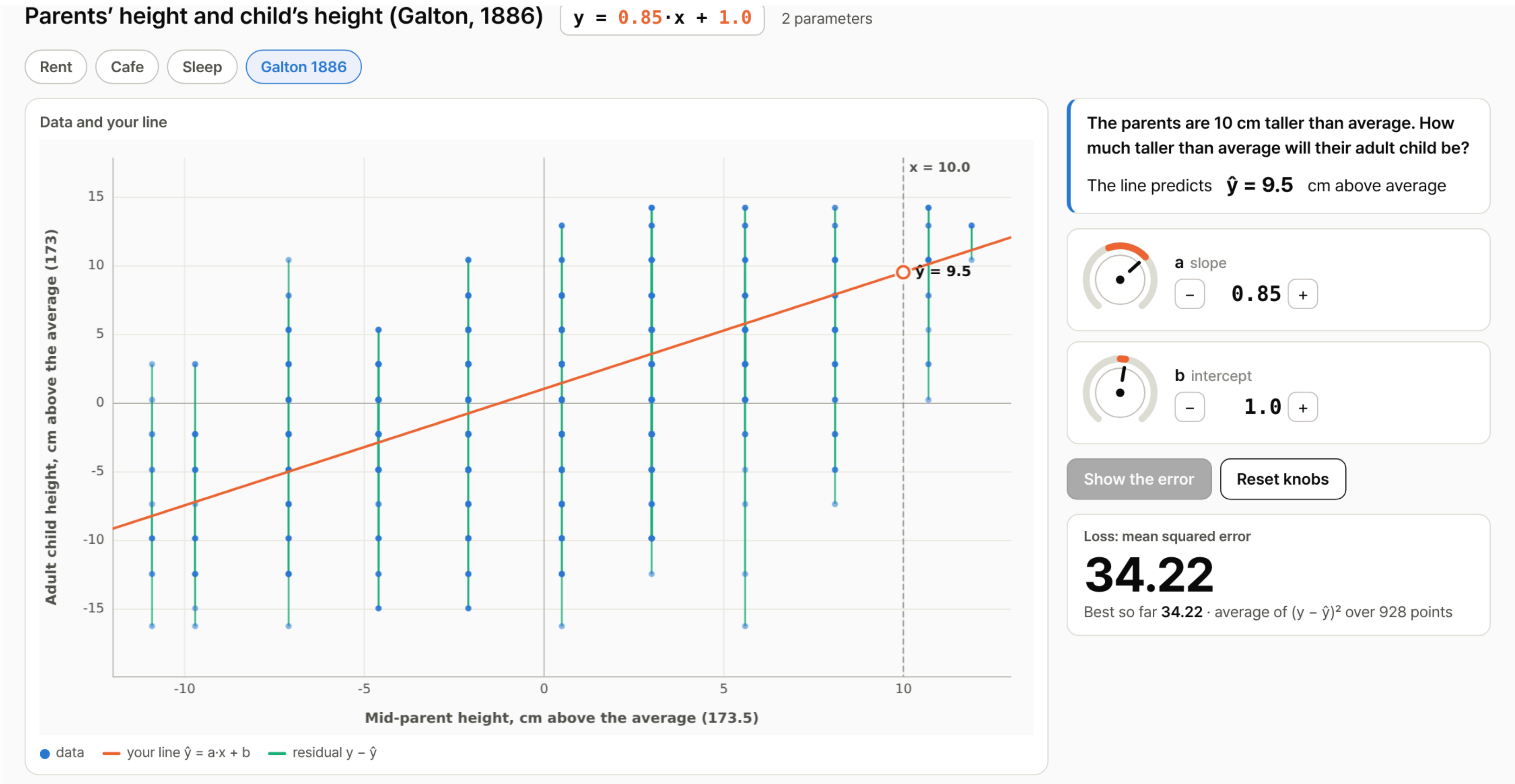
Loss on the Screen

Demo, tab 2, "Show the error"



Parents and Children

Galton (1886), 928 adult children of 205 families

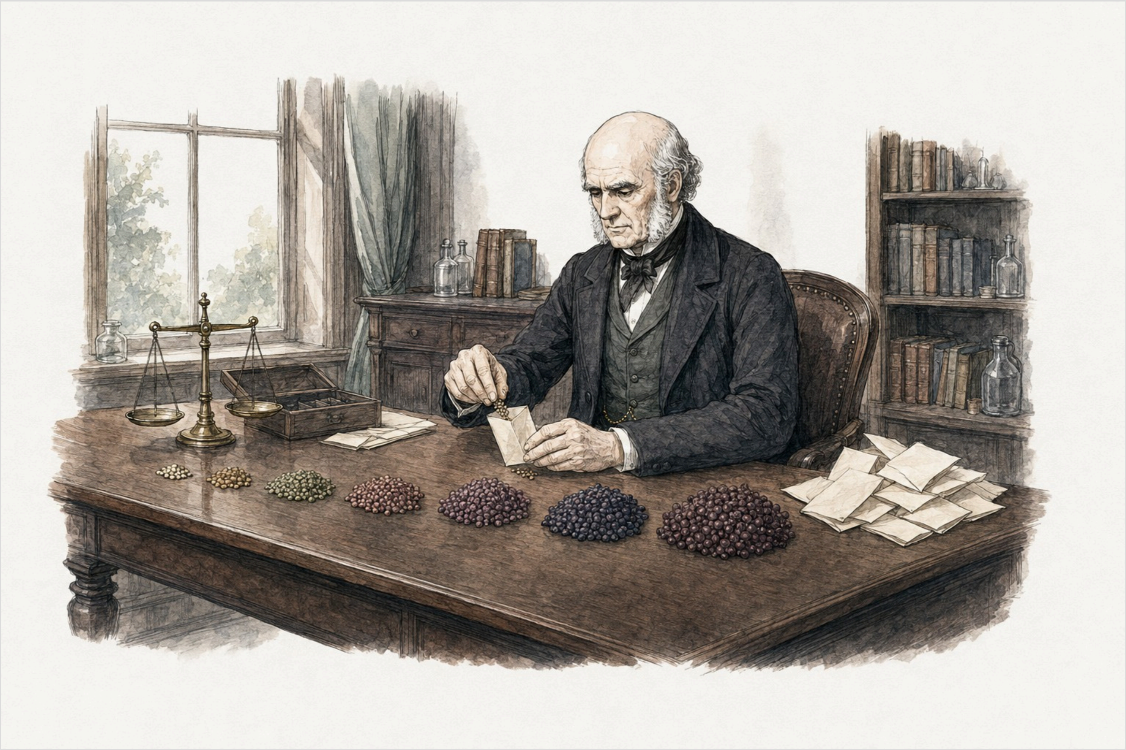


- Slope 0.65: parents 10 cm above average have children about 6.5 cm above average

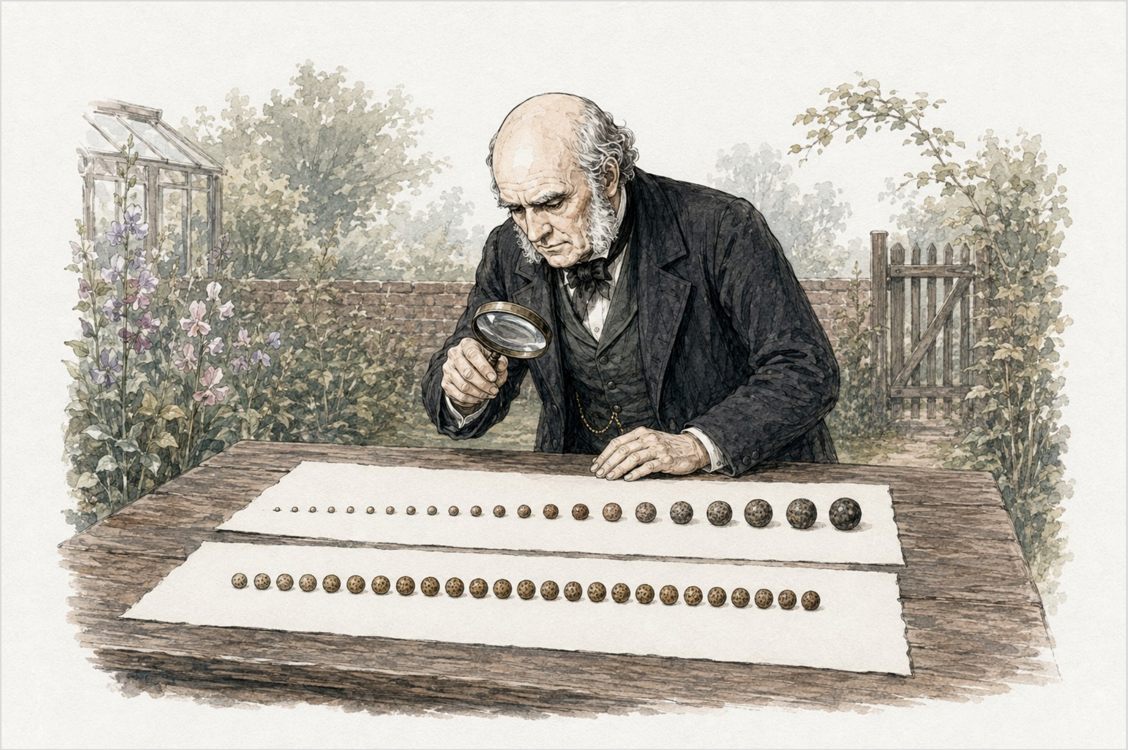
Why is it called regression?

Sweet Pea Seeds

1875



1875. Francis Galton, a cousin of Charles Darwin, sorts sweet pea seeds into seven sizes and mails packets to friends across Britain to grow.

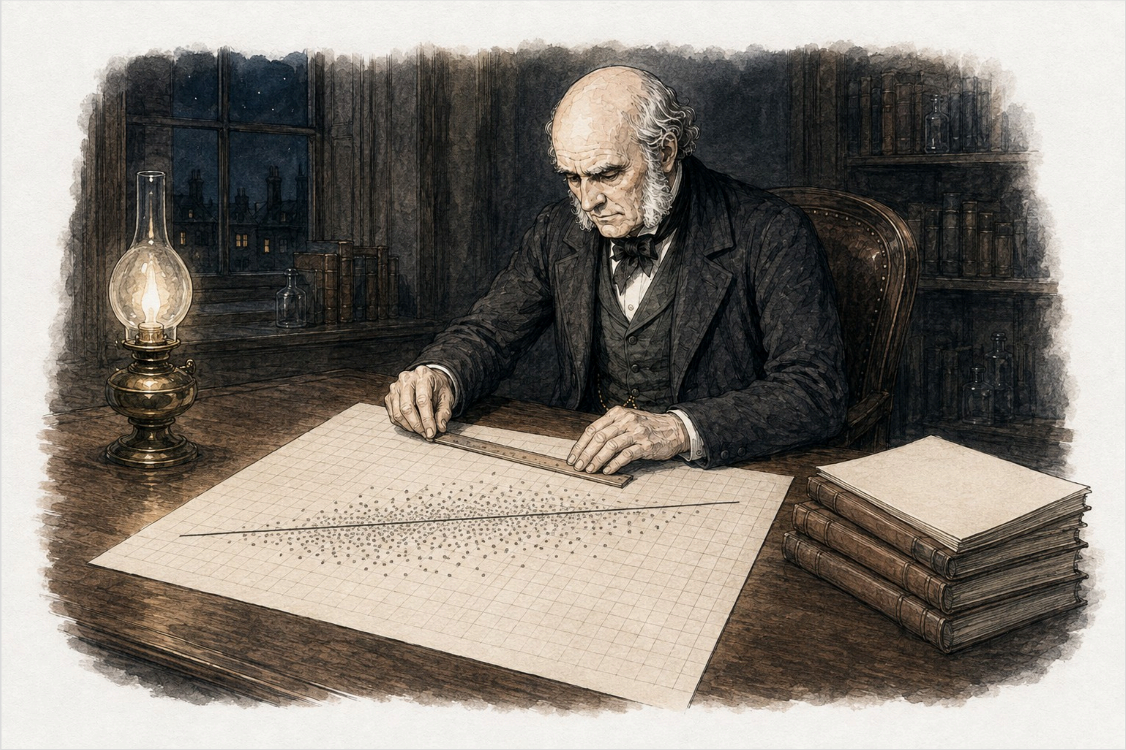


The harvest. Seeds from the largest parents come out smaller than their parents. Seeds from the smallest parents come out larger. Every group moves toward the middle size.

Measuring Heights

1884 to 1886

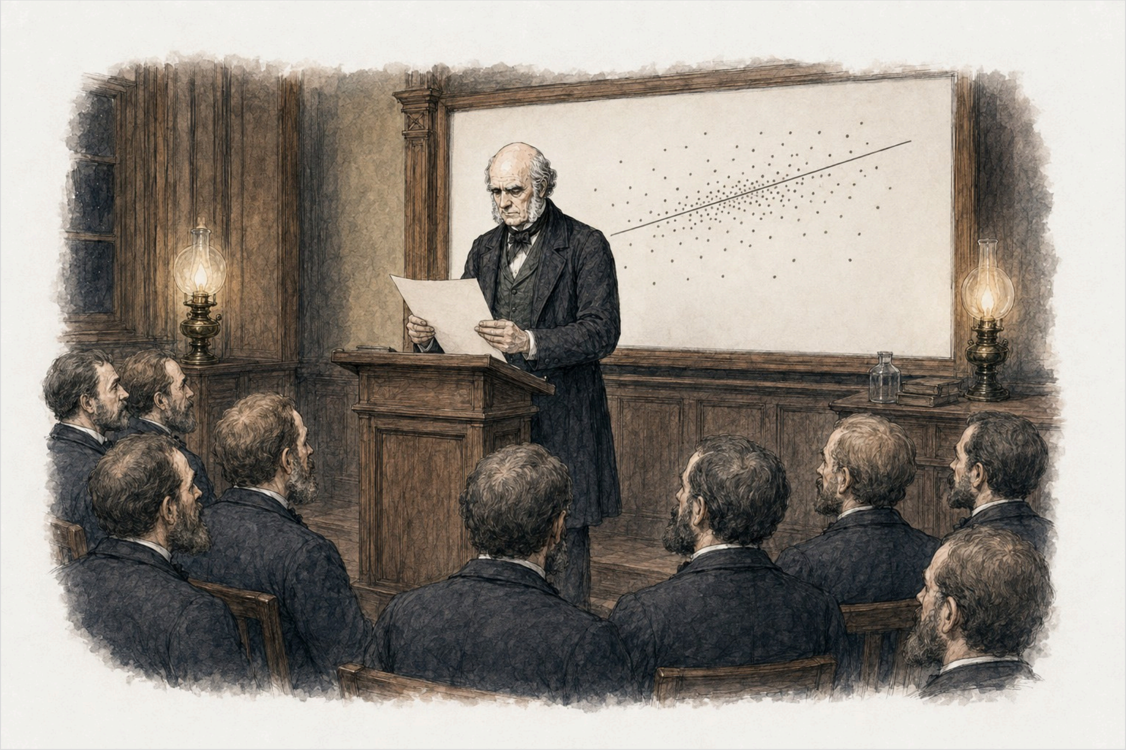
1884. At the International Health Exhibition in South Kensington, Galton's Anthropometric Laboratory measures thousands of visitors, including whole families.



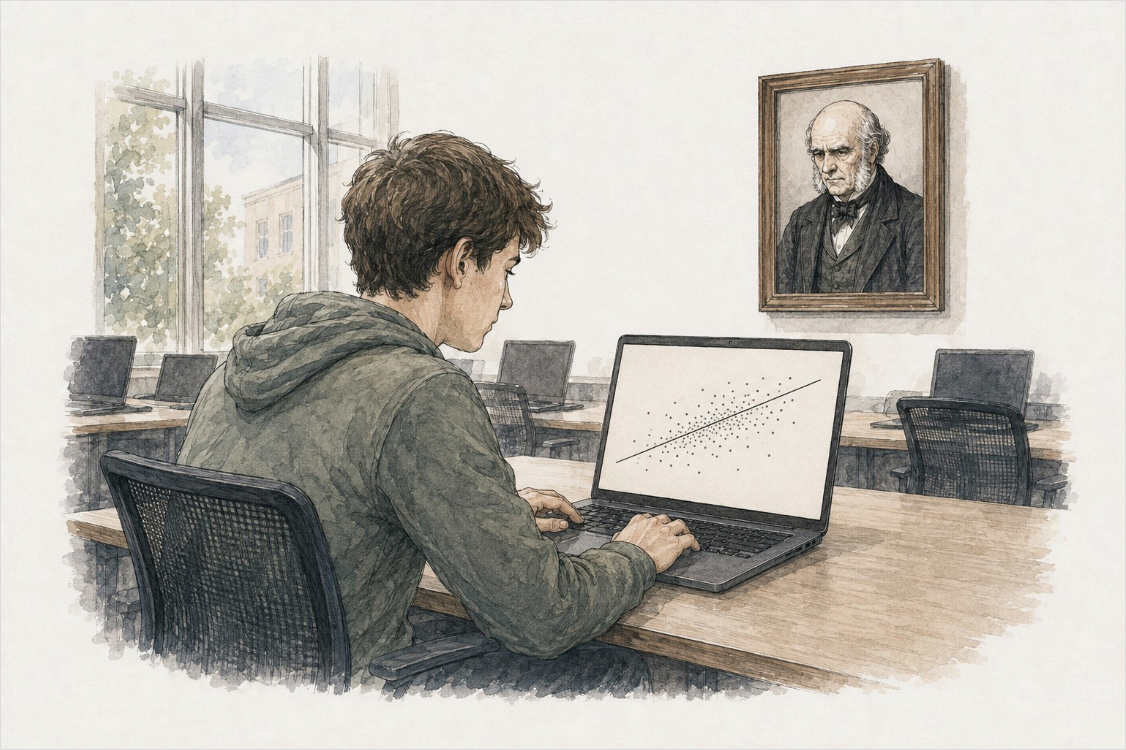
205 families, 928 adult children. Plotted against the parents' height, the children's heights follow a line with slope about two thirds.

The Word

1885 to today



1885. Galton presents "Regression towards Mediocrity in Hereditary Stature". Regression: the children's heights go back toward the average.



Today. Any model that predicts a number is called a regression model, whatever the data. The name stayed after the original meaning was dropped.

Regression Today

- Regression: a model whose output is a number (rent, cups, hours, height)
- Classification: a model whose output is a category (spam or not, which digit)
- Nothing in a regression model has to regress toward anything

Let the computer search

Brute Force

Grid search

- Choose N values for a and N values for b, evenly spaced
- Compute the loss for every one of the N × N combinations
- Keep the combination with the smallest loss

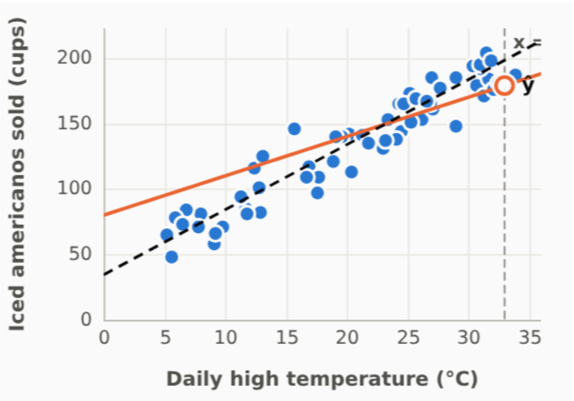
The Loss Map

Demo, tab 3, cafe data, 30 steps per parameter

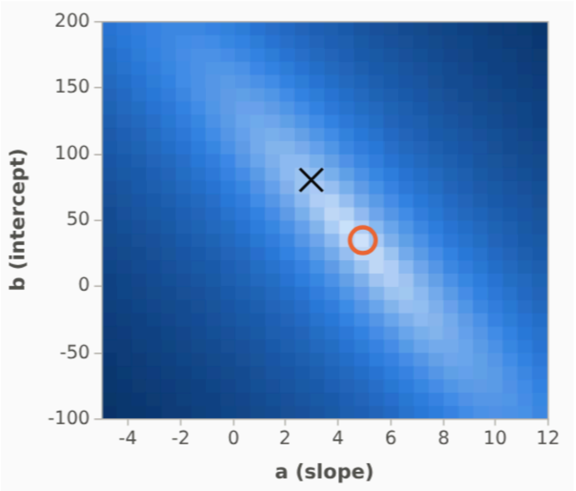
Daily high temperature and iced americano sales `try every (a, b) on a grid, keep the smallest loss`

Rent | Cafe | Sleep | Galton 1886

Data, your hand-fit line (orange), best grid line (dark)



Loss for every (a, b) on the grid. Light = small loss. × your hand fit, ○ grid best



Steps per parameter 10 | **30** | 100

Run the search | Put the best (a, b) on the knobs

Loss evaluations

900

30 × 30 = **900** combinations of (a, b)

Best on the grid: a = 5.0, b = 34, loss **174.6**
900 evaluations computed in 2 ms. The grid beat or matched your hand fit (441.5).

Your hand fit: a = 3.0, b = 80, loss **441.5**

Parameters k	Steps N	Evaluations N^k	Time at 470,000 per second
2	10	100	under 1 ms
2	**30**	**900**	**2 ms**
2	100	10,000	21 ms
2	1000	1,000,000	2.1 s

Time is projected from the speed measured on this computer during the last run.

Cost of a Grid

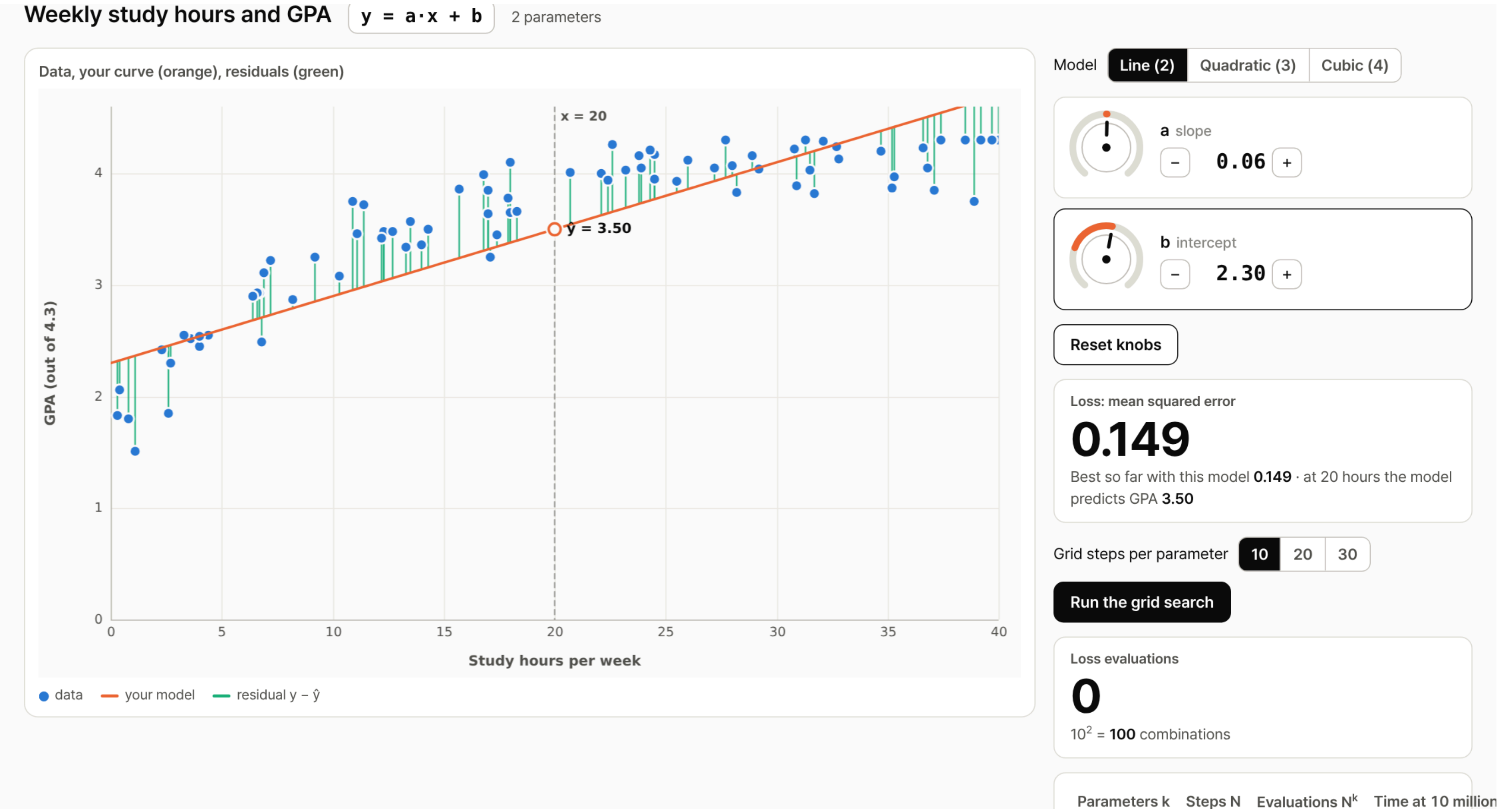
k parameters, N steps each, N^k evaluations

Parameters k	Steps N	Evaluations	Time at 3 million per second
2	100	10^4	3 ms
3	100	10^6	0.3 s
4	100	10^8	33 s
8	100	10^{16}	100 years
12	100	10^{24}	10 billion years

When a line is not enough

Study Hours and GPA

The best line still misses the shape



- Residuals are positive in the middle and negative at both ends: the data curves

Adding Terms

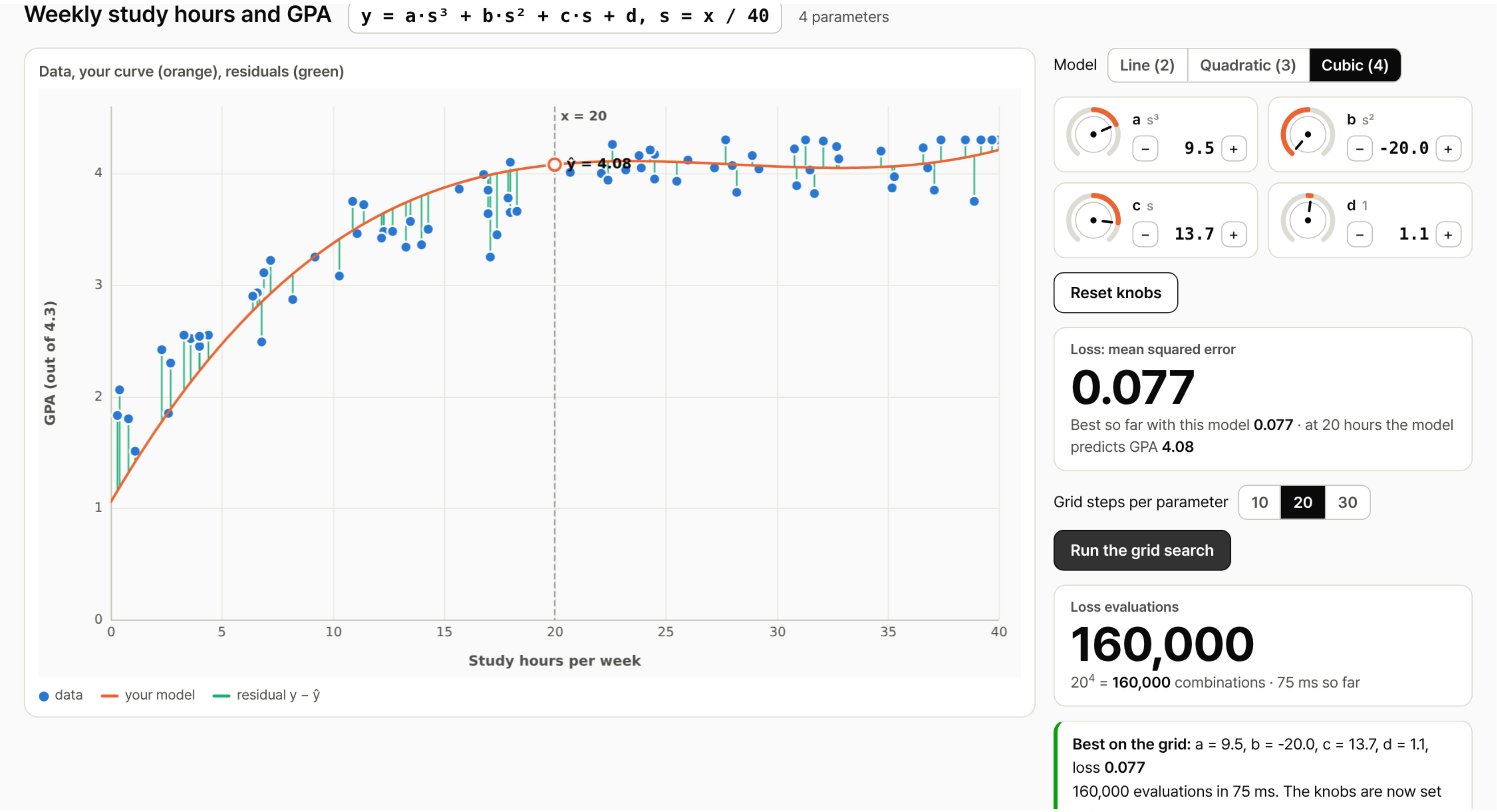
More parameters, more shapes

Model	Equation	Parameters
Line	$y = a \cdot s + b$	2
Quadratic	$y = a \cdot s^2 + b \cdot s + c$	3
Cubic	$y = a \cdot s^3 + b \cdot s^2 + c \cdot s + d$	4

- $s = x / 40$, so that every term stays in a similar range

Cubic by Grid Search

Demo, tab 4, 20 steps per parameter



- 20^4 = 160,000 evaluations in under 0.1 s. 100^4 would take half a minute

A function with everything in it

The Mystery Function

Eight weights, all hidden

$$y = w_1 \sin(2x) + w_2 \cos(3x) + w_3\, e^x/20 + w_4 \ln(x+4) + w_5\, (x/3)^4 + w_6\, (x/3)^2 + w_7\, (x/3) + w_8$$

- 120 data points are given, with a little noise
- Each weight is somewhere between –5 and 5

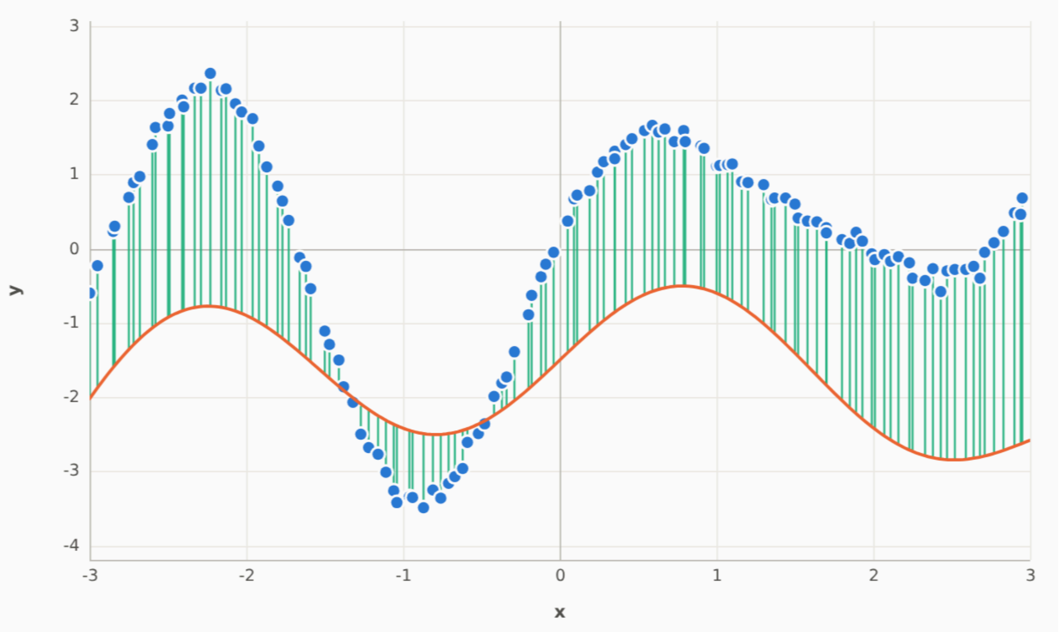
By Hand

Demo, tab 5

A function with everything in it 8 parameters. The true weights are hidden.

$y = w_1 \sin(2x) + w_2 \cos(3x) + w_3 e^x/20 + w_4 \ln(x+4) + w_5 (x/3)^4 + w_6 (x/3)^2 + w_7 (x/3) + w_8$

Data, your curve (orange), residuals (green)



data · your model · residual $y - \hat{y}$

w_1 sin(2x) − **1.00** +

w_2 cos(3x) − **0.00** +

w_3 $e^x/20$ − **0.00** +

w_4 ln(x+4) − **0.00** +

w_5 $(x/3)^4$ − **-0.80** +

w_6 $(x/3)^2$ − **0.00** +

w_7 x/3 − **0.00** +

w_8 1 − **-1.50** +

Reset knobs

Loss: mean squared error

4.002

Best so far **1.089** · the noise alone is about 0.006

Cost of a grid search for this model

Parameters k	Steps N	Evaluations N^k	Time at 2.1 million per second
8	10	100,000,000	46.9 s
8	30	6.6×10^{11}	3.6 days
8	100	1.0×10^{16}	149 years

By Grid

8 parameters

Steps N	Evaluations	Time at 3 million per second
10	10^8	30 s, but the grid points are 1 apart
30	6.6×10^{11}	2 days
100	10^{16}	100 years

Machine learning

Three Parts of Machine Learning

- A model: a function with parameters, $y = f(x; w)$
- A loss: one number that measures how far the predictions are from the data
- A search: a procedure that lowers the loss without trying every combination

Next Time

Gradient descent

- The machine turns each knob a small step in the direction that lowers the loss, and repeats
- The same procedure works for 2 parameters and for 2 billion

What if the machine could turn the knobs by itself?